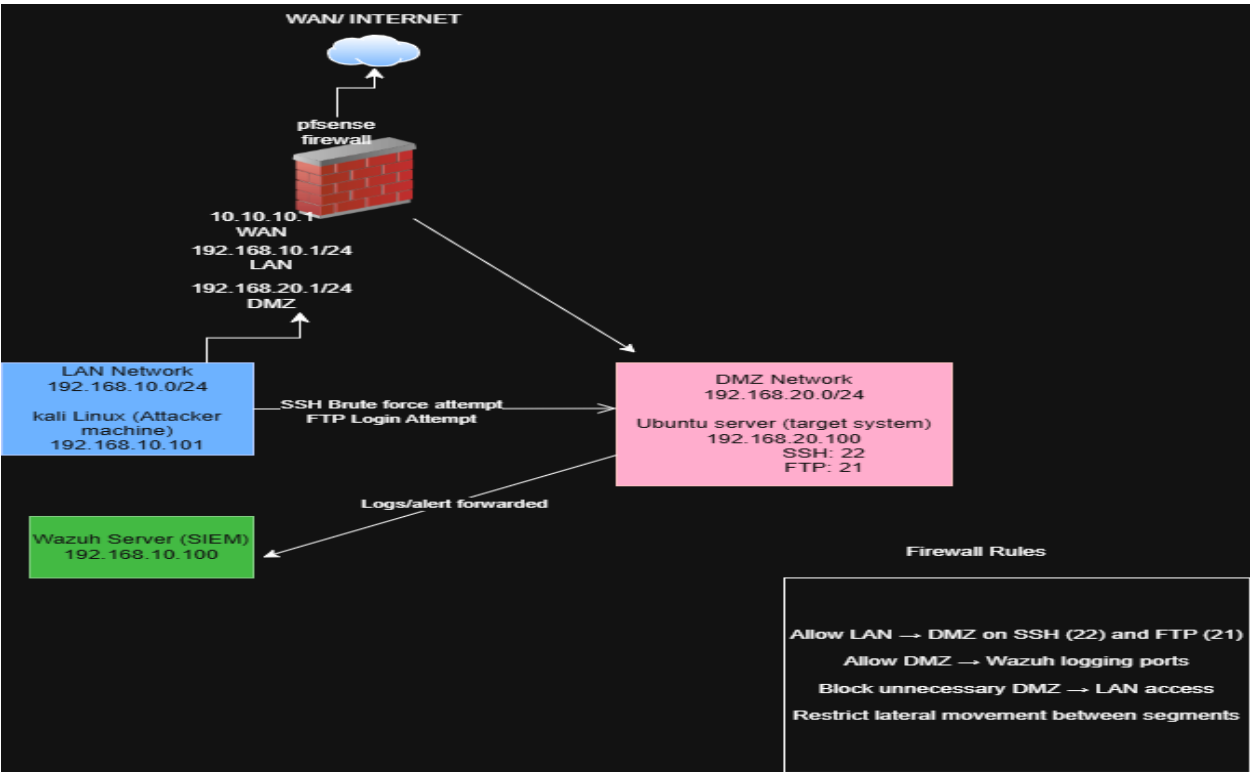


Network Security Monitoring and Intrusion Detection using Wazuh

This report demonstrates a simulated network security monitoring environment using pfSense firewall, Kali Linux attacker machine, Ubuntu server (target), and Wazuh SIEM for threat detection and log analysis. The environment is divided into LAN and DMZ network segments and monitored through centralized security logging.

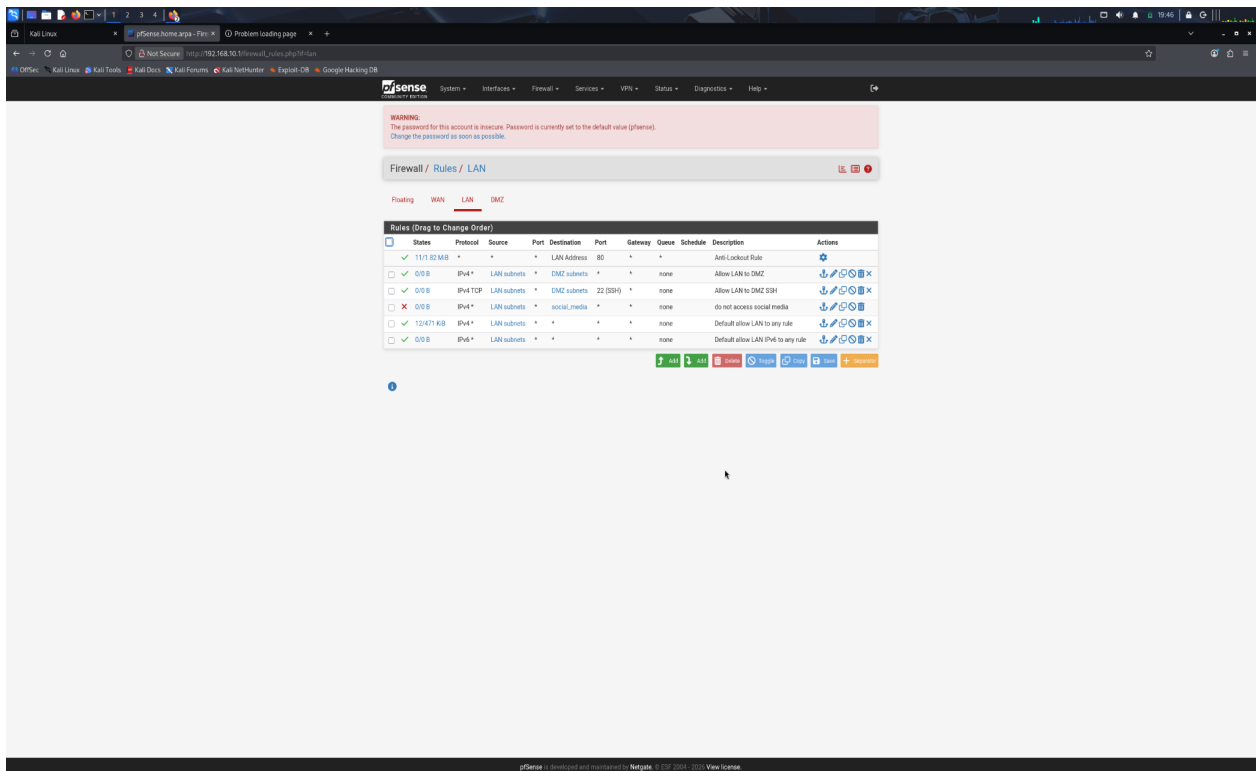
Network Diagram

Network architecture showing WAN/Internet, pfSense firewall, LAN attacker machine, DMZ Ubuntu server target, and Wazuh SIEM server.



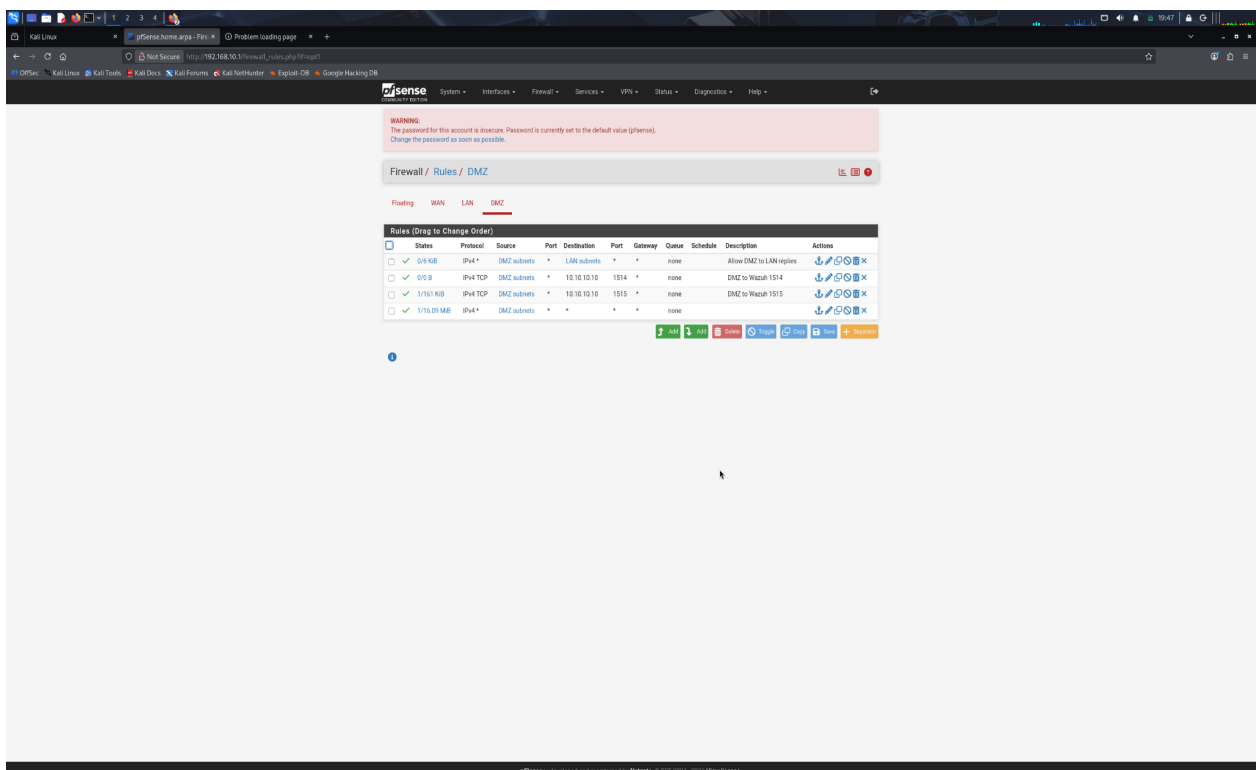
pfSense LAN Firewall Rules

Firewall rules configured on the LAN interface allowing controlled communication to the DMZ.



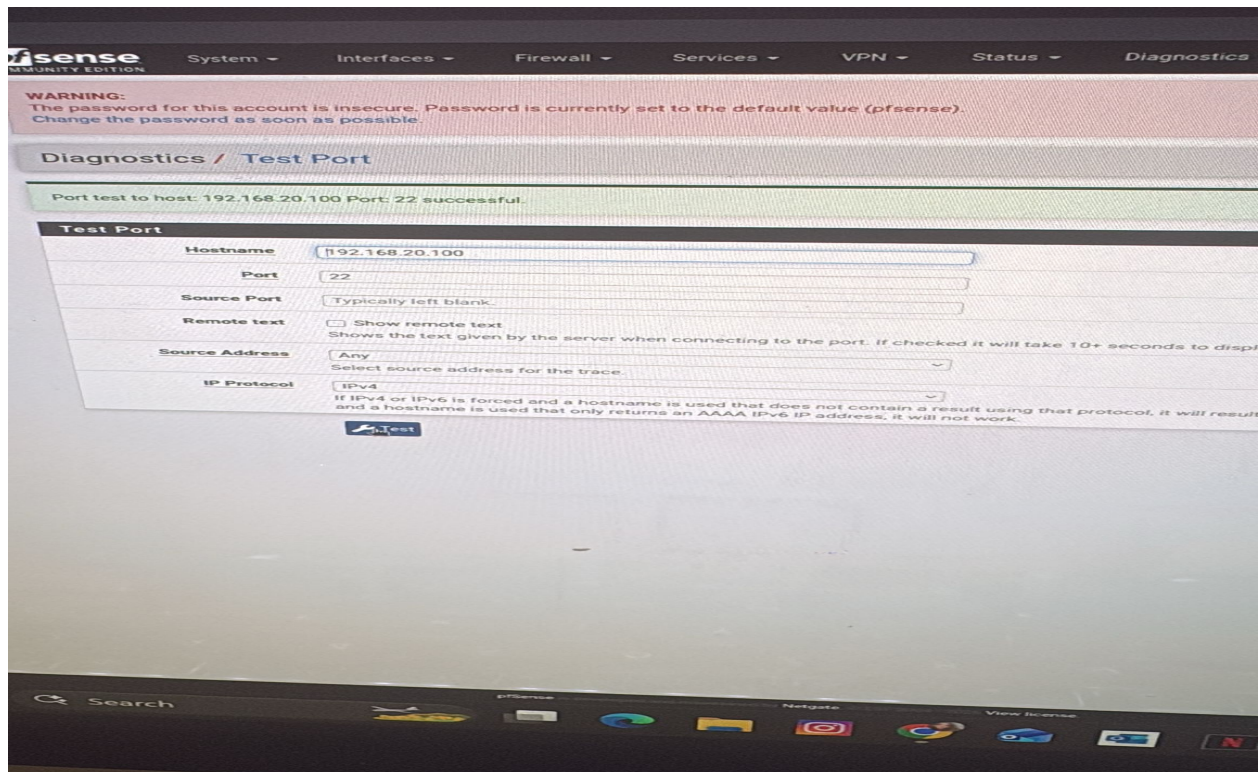
pfSense DMZ Firewall Rules

Firewall rules configured on the DMZ interface controlling traffic and logging communication to Wazuh.



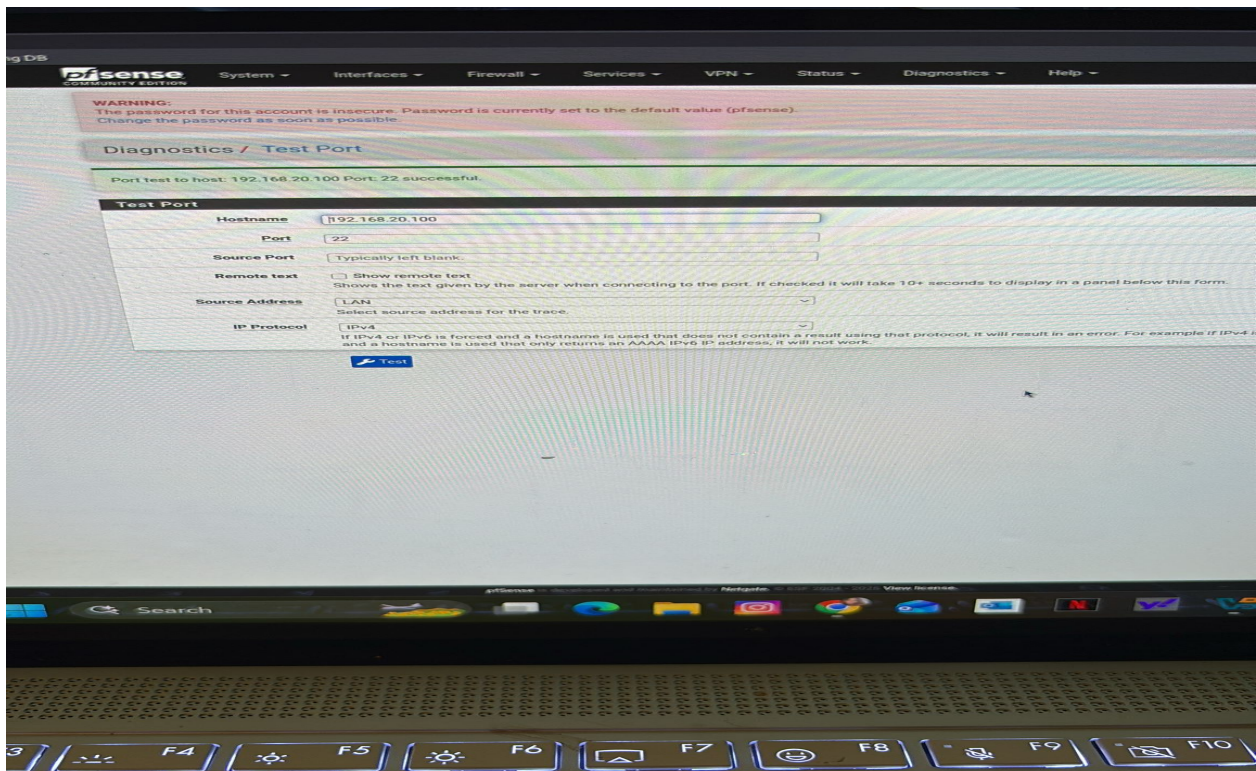
Port Connectivity Test

pfSense diagnostic test confirming connectivity to the Ubuntu server SSH port (22).



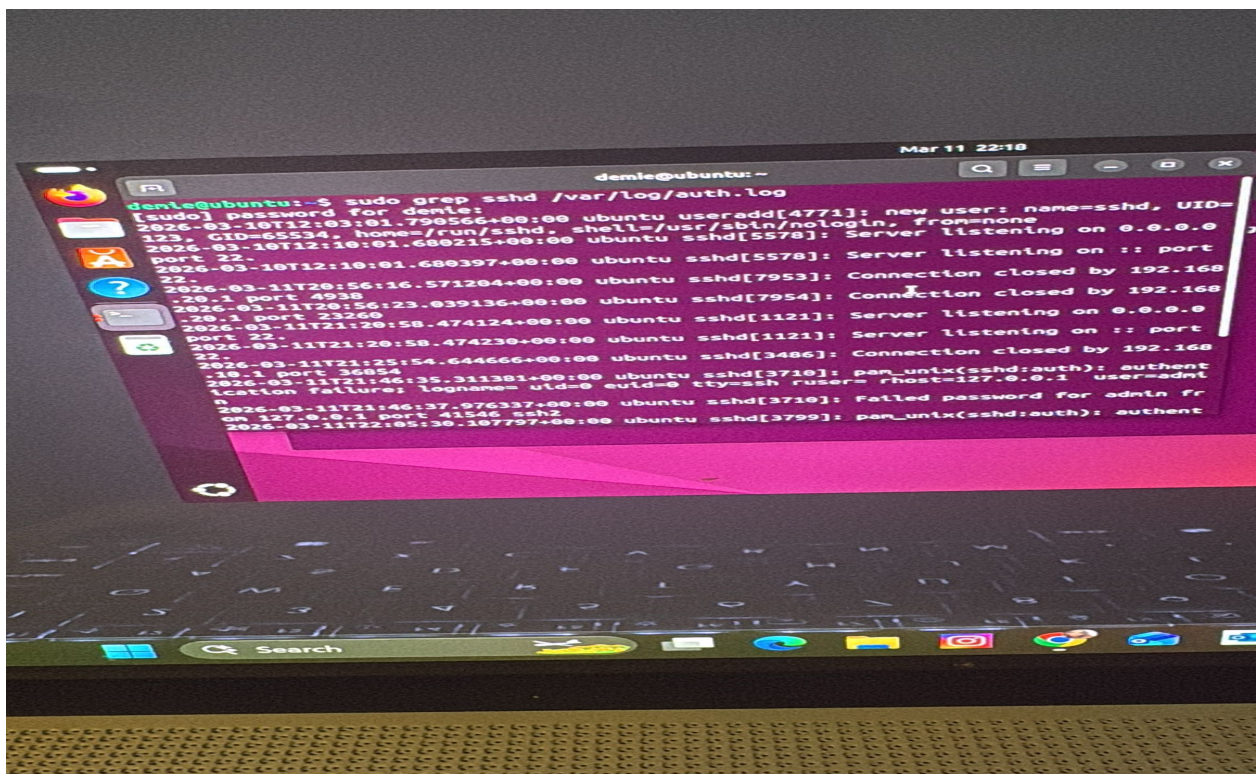
Port Connectivity Verification

Additional connectivity verification confirming successful communication with the Ubuntu server.



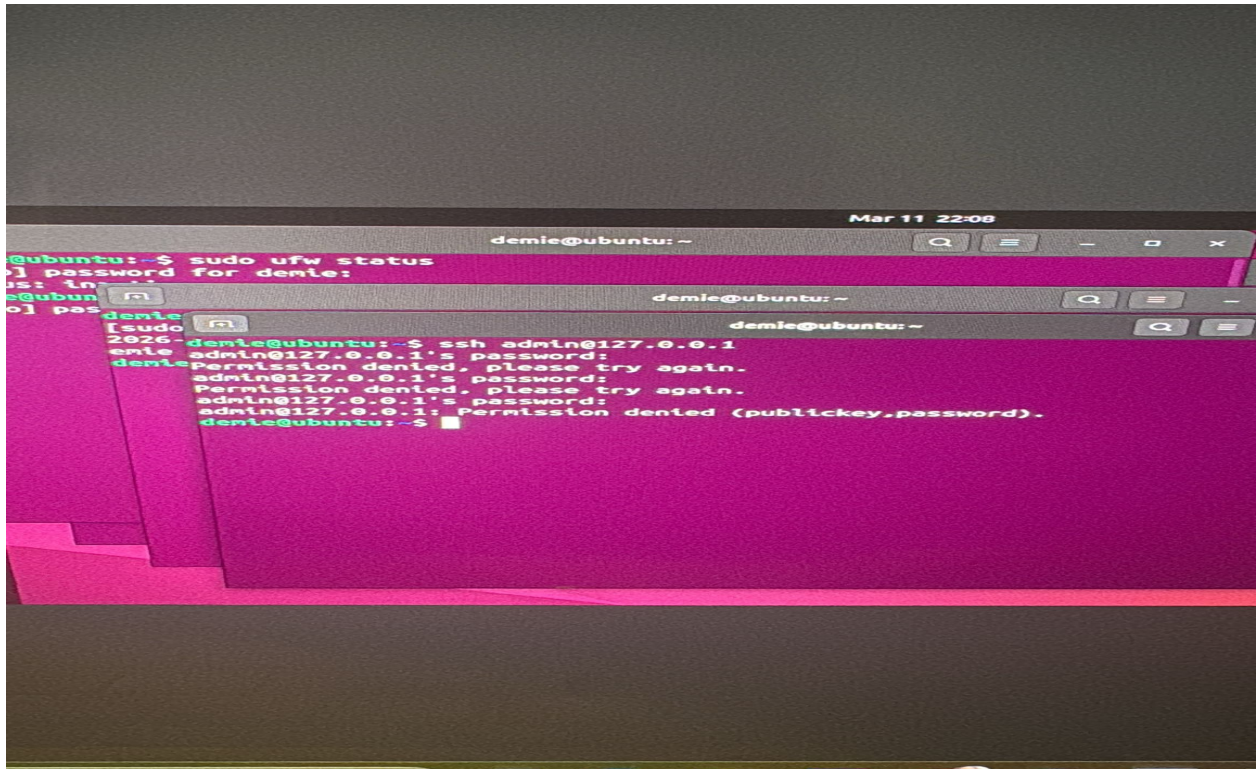
SSH Failed Login Attempt

Terminal evidence showing failed SSH authentication attempts during the simulated attack.



Log File Evidence

System authentication logs showing SSH login attempts captured in /var/log/auth.log.



Threat Detection Summary

During the simulation, an attacker machine (Kali Linux) attempted multiple SSH login attempts against the Ubuntu server located in the DMZ network. These attempts were intentionally configured to fail in order to generate authentication errors. The Ubuntu server recorded these attempts in the authentication logs located in /var/log/auth.log. The log entries clearly show repeated failed password attempts for the user 'admin', which indicates a possible brute-force attack. These logs were forwarded to the Wazuh SIEM server where they can be monitored and analyzed for suspicious activity. Security analysts can use this information to detect unauthorized access attempts and respond to threats in real time.

Security Recommendations

- Implement strong password policies and avoid default credentials.
- Configure account lockout policies after multiple failed login attempts to prevent brute-force attacks.

- Restrict SSH access using firewall rules and allow connections only from trusted IP addresses.
- Enable multi-factor authentication (MFA) for administrative access.
- Continuously monitor logs using SIEM solutions such as Wazuh to detect suspicious activities.
- Regularly update operating systems and security tools to patch vulnerabilities.
- Segment networks (LAN and DMZ) to reduce the impact of potential security breaches.